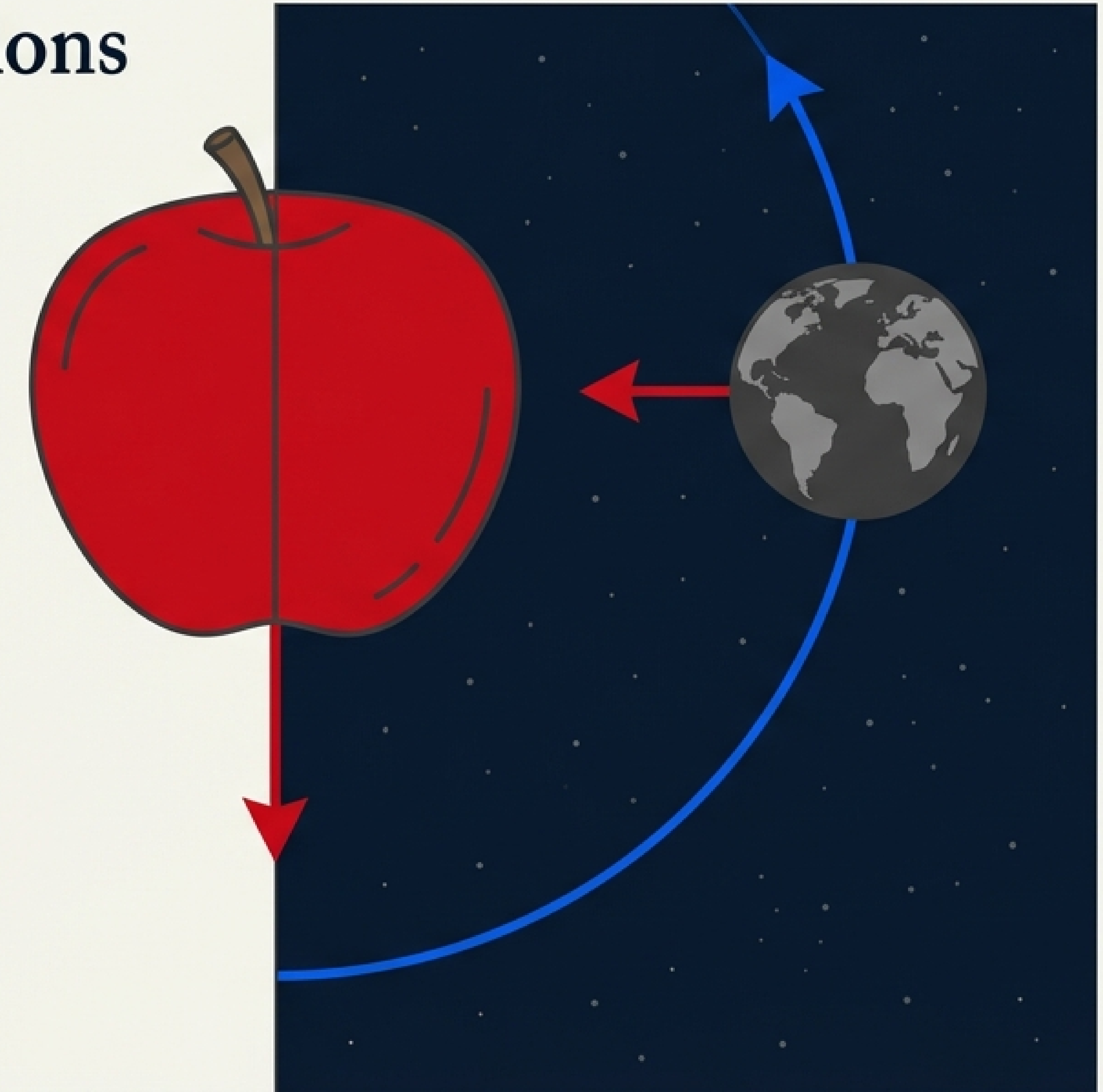
Gravitation: The Universal Bond

From Kepler's Observations to Satellite Mechanics

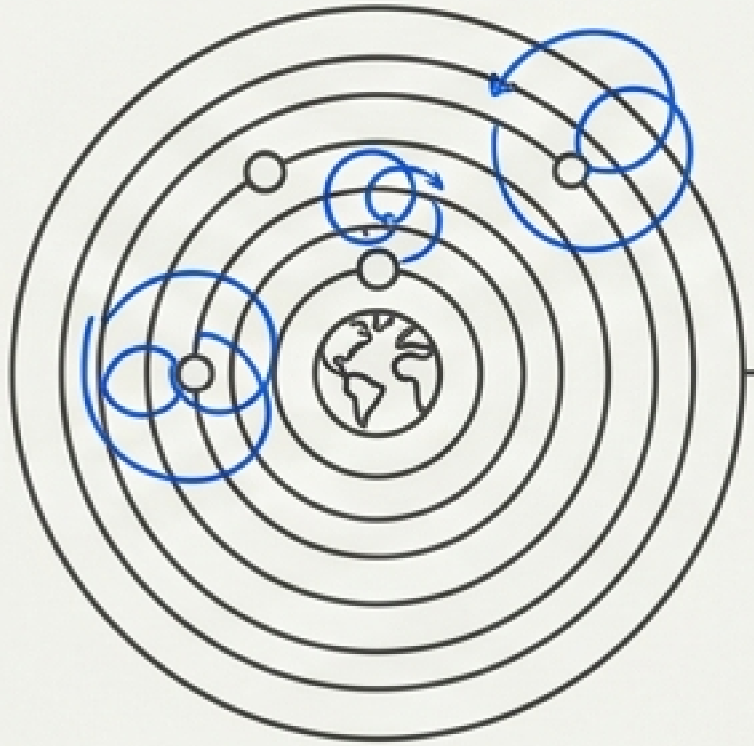


Historically, terrestrial gravity and planetary motion were seen as separate motion were seen as separate phenomena. This document traces the intellectual journey—from the earliest geometric models of Ptolemy to Newton's grand synthesis—revealing the mathematical code that governs everything from a falling stone to the structure of galaxies.

Context: Chapter 7. A comprehensive guide to the laws of motion, the variation of acceleration, and the mechanics of spaceflight.

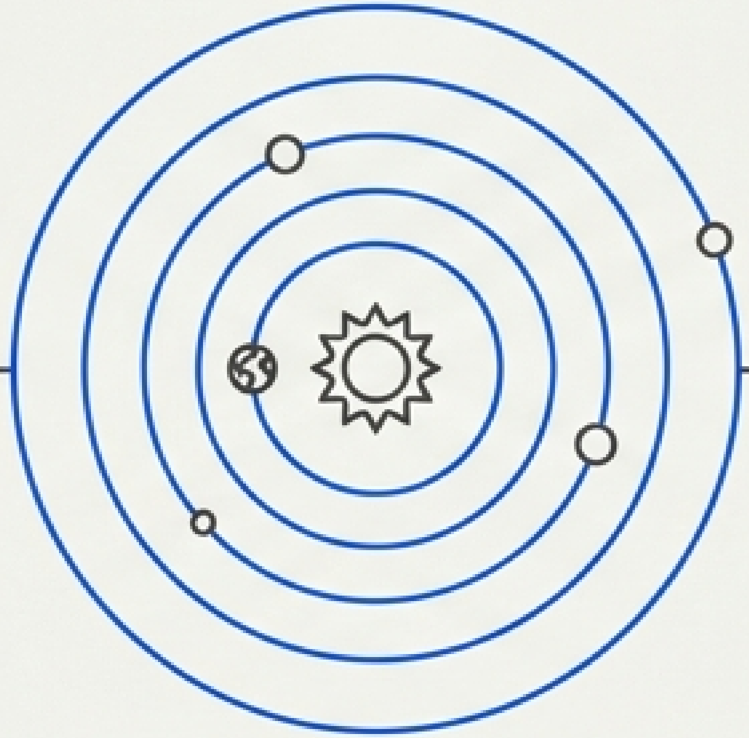
A History of Looking Up

The Geocentric Model (Ptolemy, 2000 Years Ago)



Earth-centered. Required complicated "epicycles" to explain retrograde motion.

The Heliocentric Model (Copernicus, 1543)



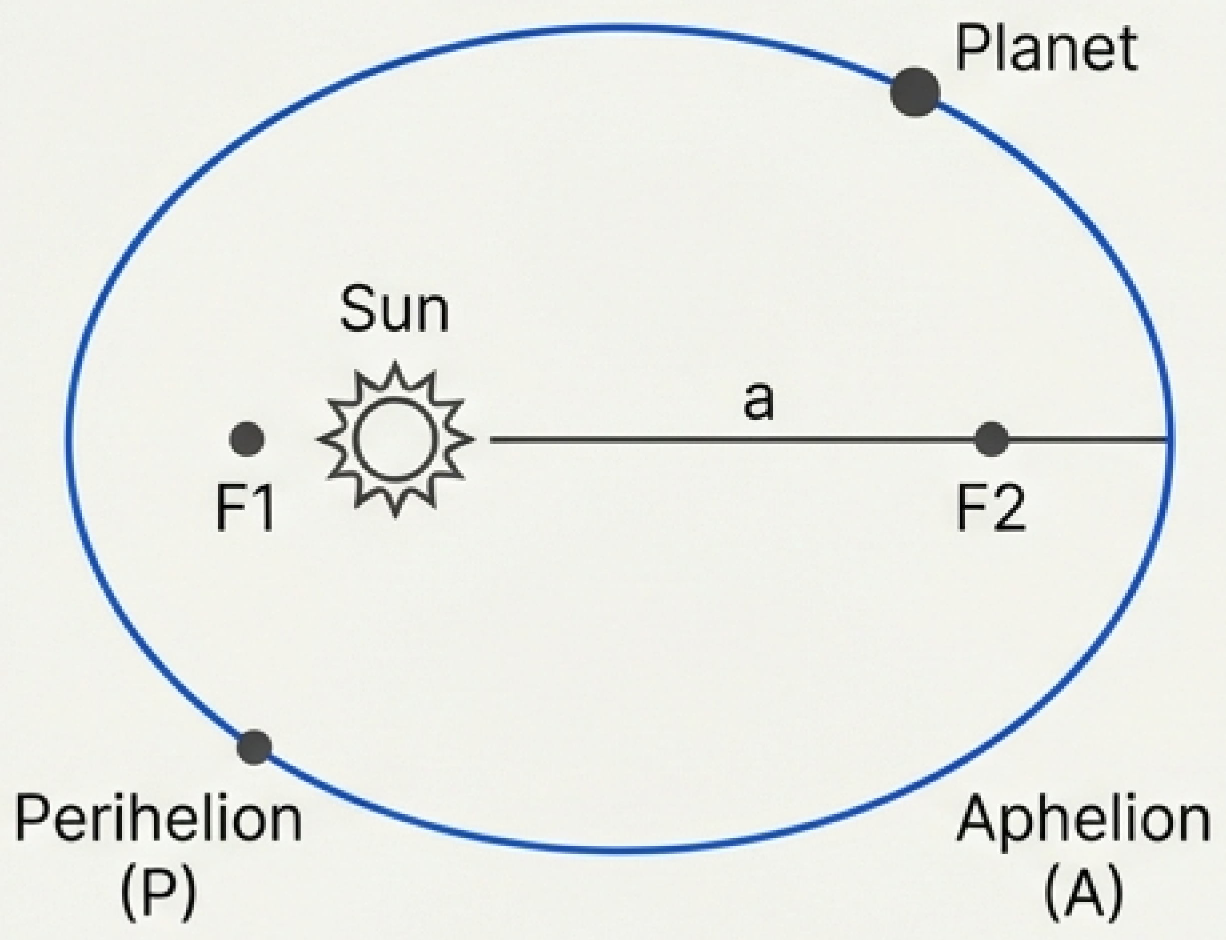
Sun-centered. A more elegant model, though originally controversial.

The Data Gatherer (Tycho Brahe, 1546-1601)

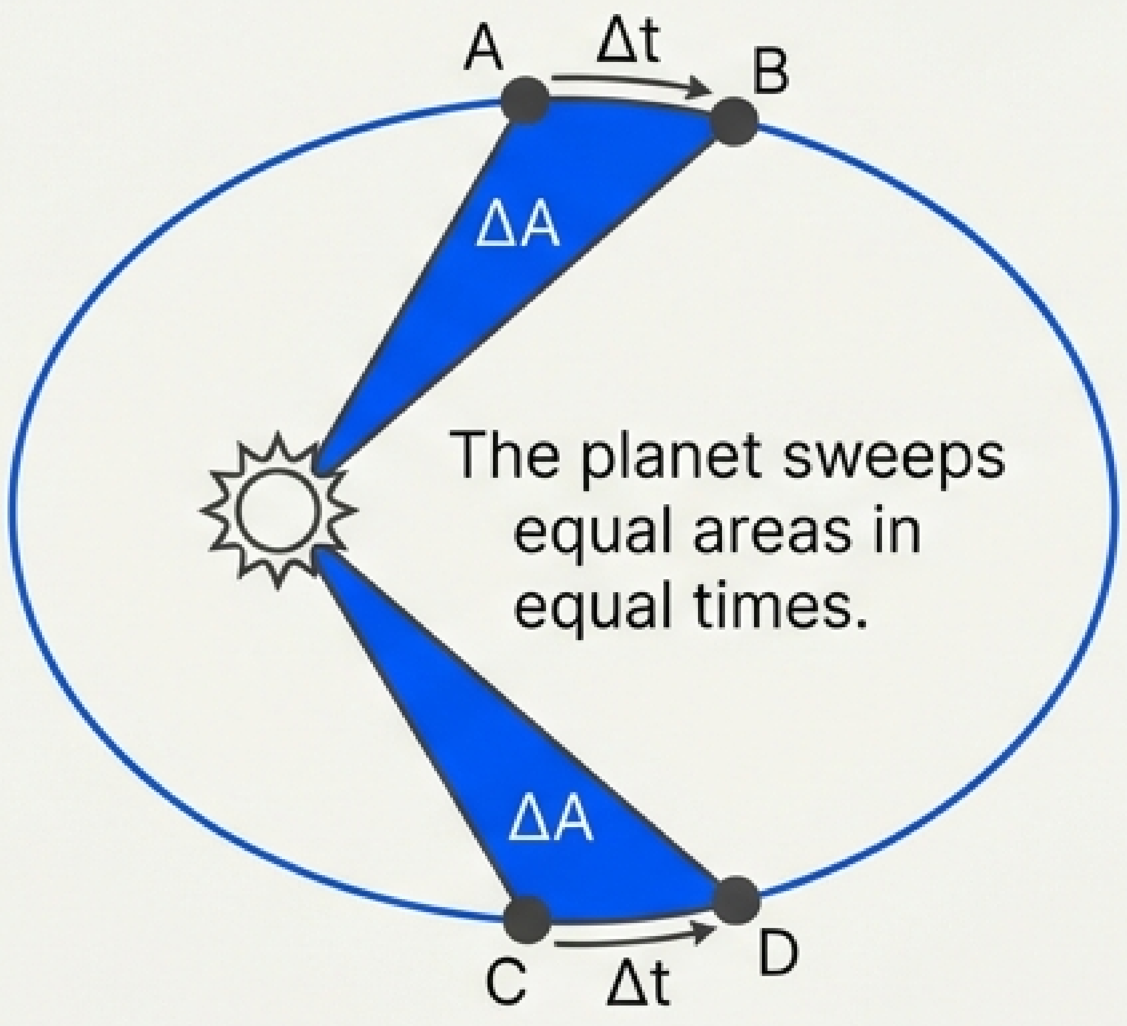
Tycho Brahe recorded massive datasets of planetary positions. This raw data allowed Kepler to unlock the true nature of orbits.

Kepler's Laws of Orbit and Areas

Law 1: Law of Orbits



Law 2: Law of Areas



Derivation: Conservation of Angular Momentum (L)

Since the gravitational force is central (along vector r), torque is zero, and L is constant.

Formula:
$\Delta A/\Delta t = L/(2m) =$ constant.

Kepler's Third Law: The Law of Periods

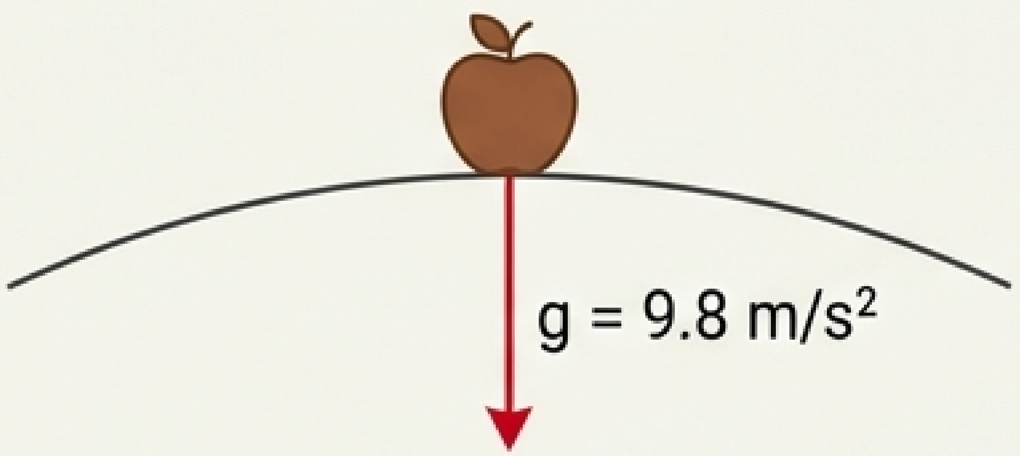
Planet	Semi-major Axis a (10^{10} m)	Period T (years)	Quotient Q (T^2/a^3)
Mercury	5.79	0.24	2.95
Earth	15.0	1.00	2.96
Mars	22.8	1.88	2.98
Saturn	143	29.5	2.98

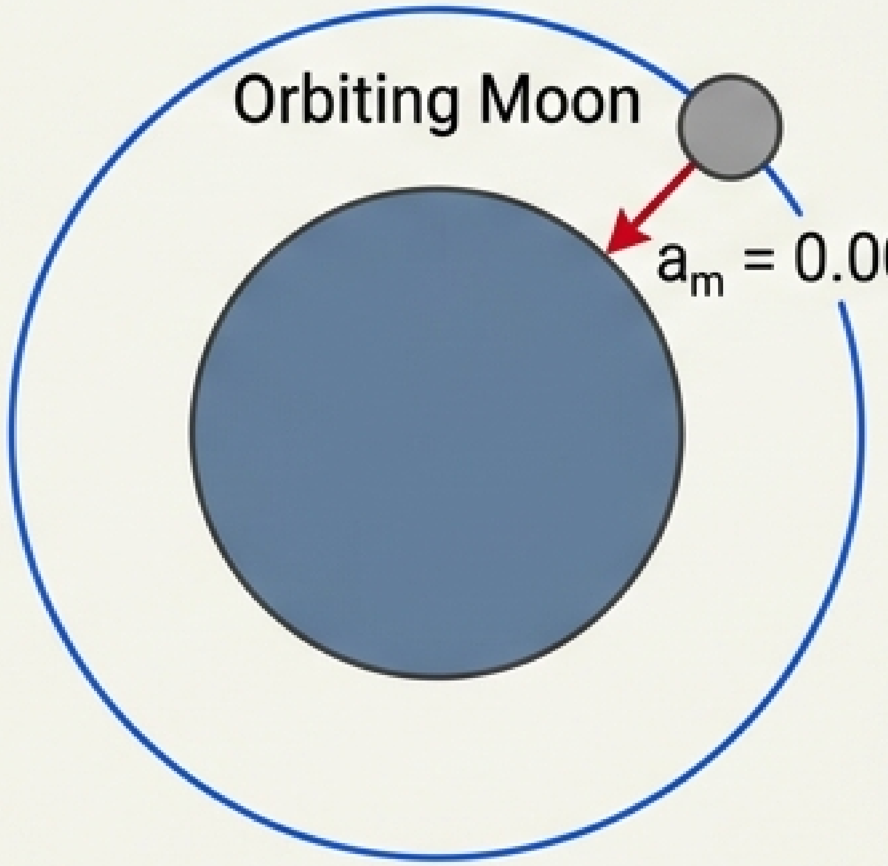
The Law: The square of the time period (T) is proportional to the cube of the semi-major axis (a).

Mathematical Harmony: $T^2 \propto a^3$.

Newton's Universal Law of Gravitation

The Inverse Square Law





Newton realized the force pulling the apple is the same force holding the Moon. The Moon is simply "falling" around the Earth.

$$F = G\frac{m_1 m_2}{r^2}$$

Derivation: Central Force & Acceleration

Newton combined Kepler's laws with his own second law of motion.

For a circular orbit, centripetal acceleration is $a_c = v^2/r$.

The gravitational force provides this acceleration, so $F = ma_c$.

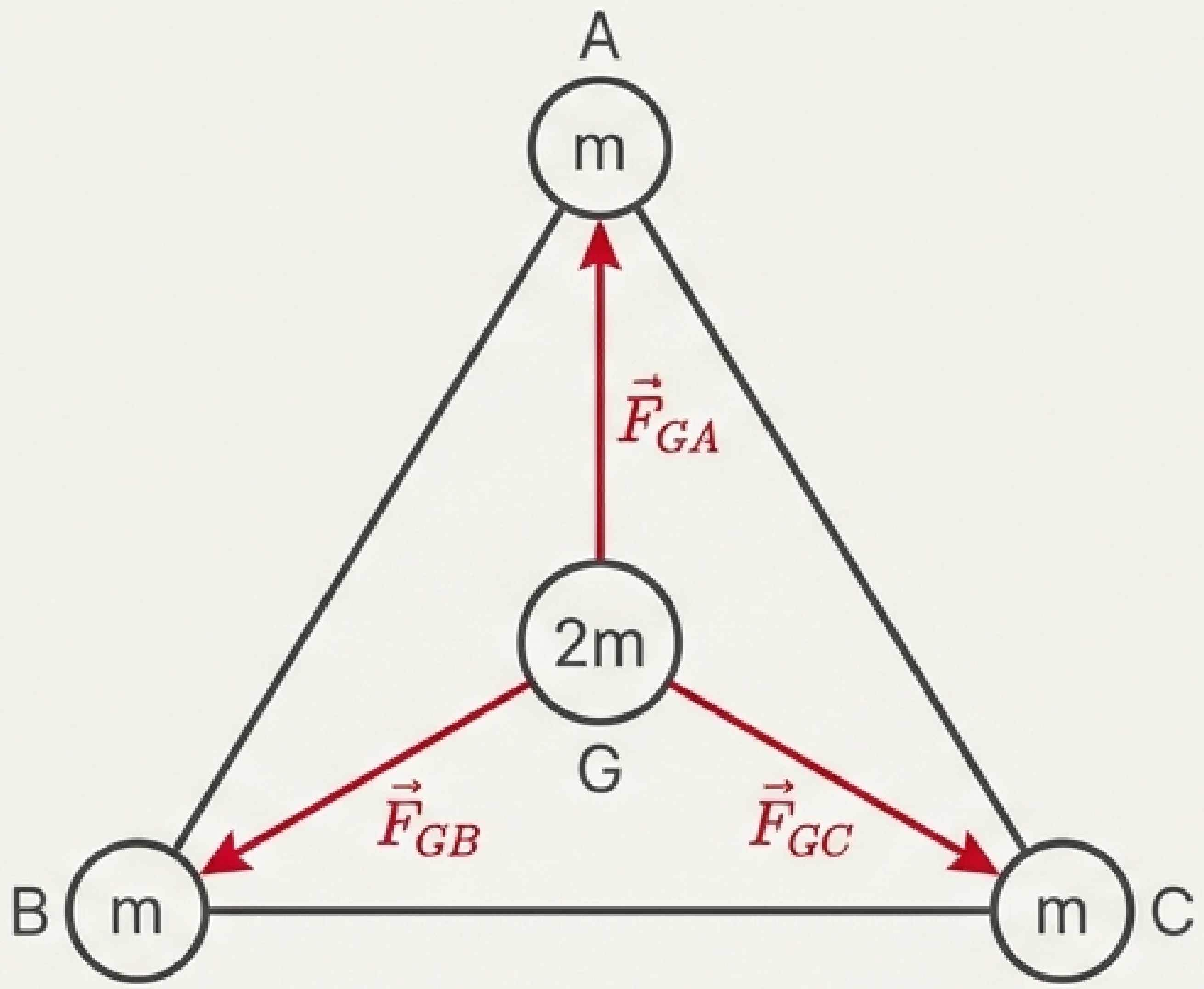
Since $v = 2\pi r/T$, substituting gives $F \propto r/T^2$.

Using Kepler's Third Law, $T^2 \propto r^3$, we get

$F \propto r/(r^3) \propto 1/r^2$.

Note: G is the gravitational constant, determined later by Cavendish.

The Vector Nature of Gravity



Principle of Superposition: The force on a mass due to a collection of other masses is the vector sum of the individual forces.

$$\vec{F}_R = \vec{F}_1 + \vec{F}_2 + \ldots + \vec{F}_n$$

Symmetry & Net Force

Due to the symmetry of the equilateral triangle and the identical masses at the vertices, the magnitudes of the forces exerted on the centroid mass (2m) are equal:

$$|\vec{F}_{GA}| = |\vec{F}_{GB}| = |\vec{F}_{GC}| = \frac{G(2m)m}{r^2},$$

where r is the distance from the centroid to each vertex.

Vector Summation

The vector sum of the forces is:

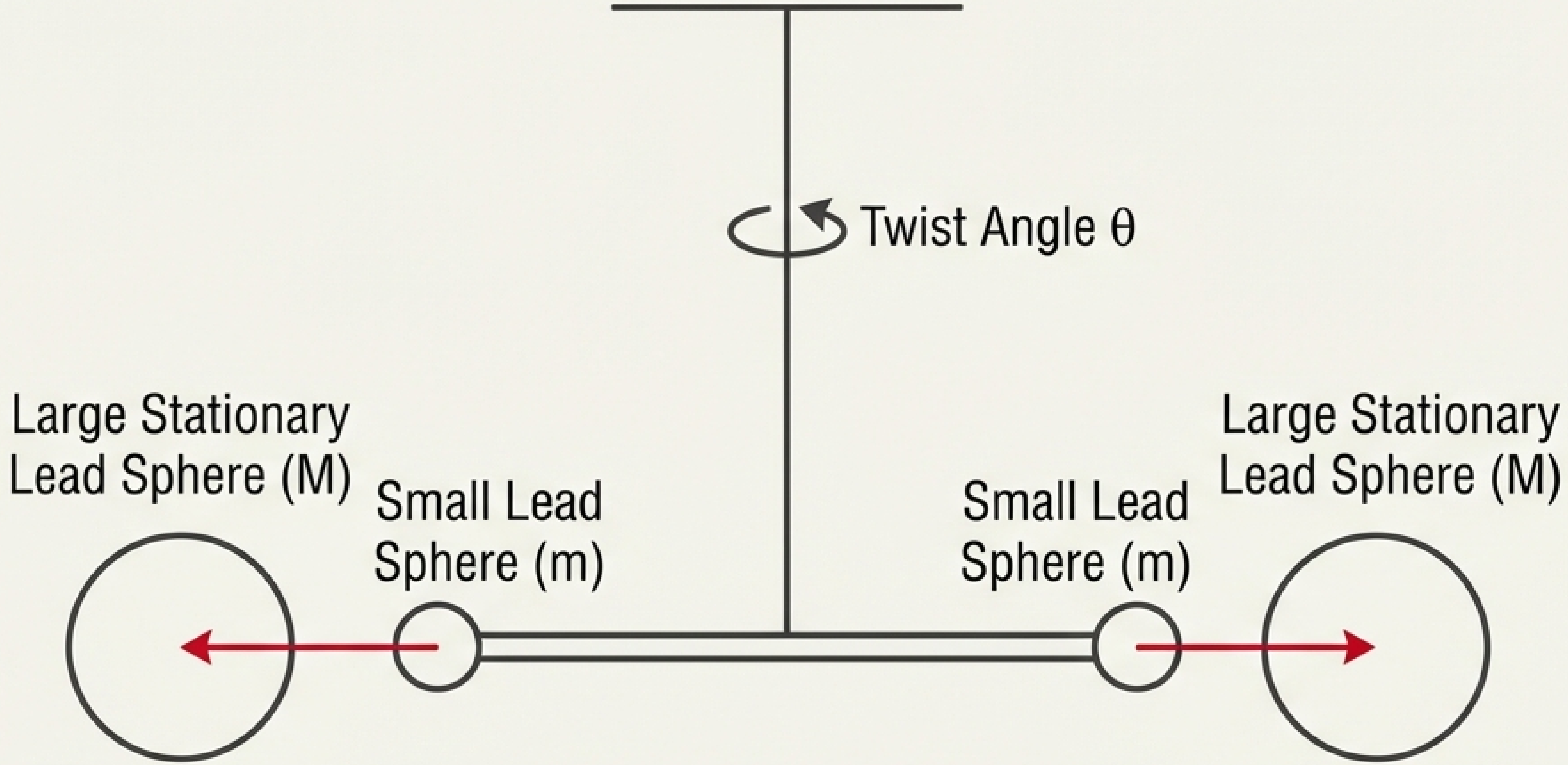
$$\vec{F}_{net} = \vec{F}_{GA} + \vec{F}_{GB} + \vec{F}_{GC}$$

By resolving these vectors into components or using geometric arguments, it can be shown that their sum is zero:

$$\vec{F}_{net} = 0$$

The net gravitational force on the mass at the centroid is zero.

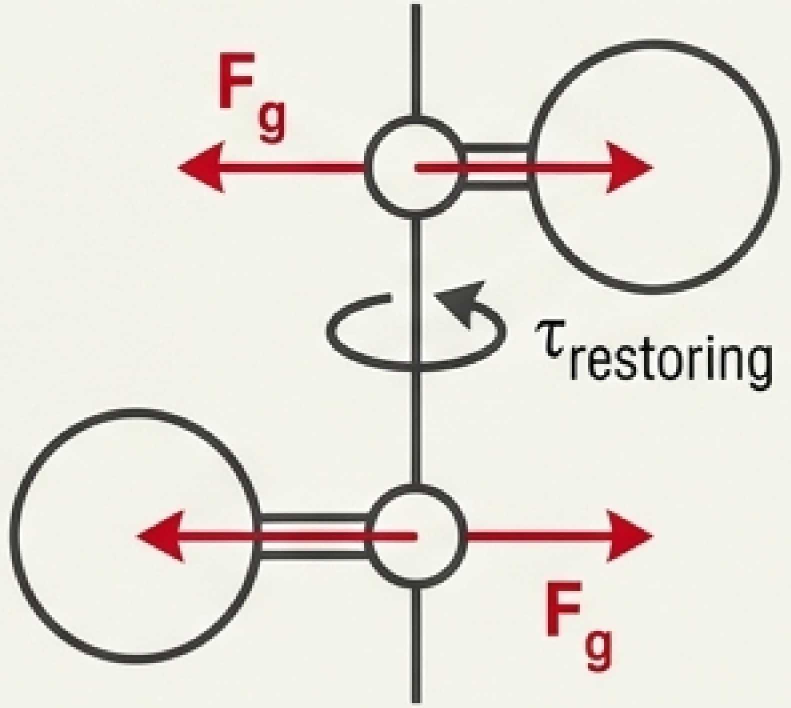
The Elusive Constant: Finding G



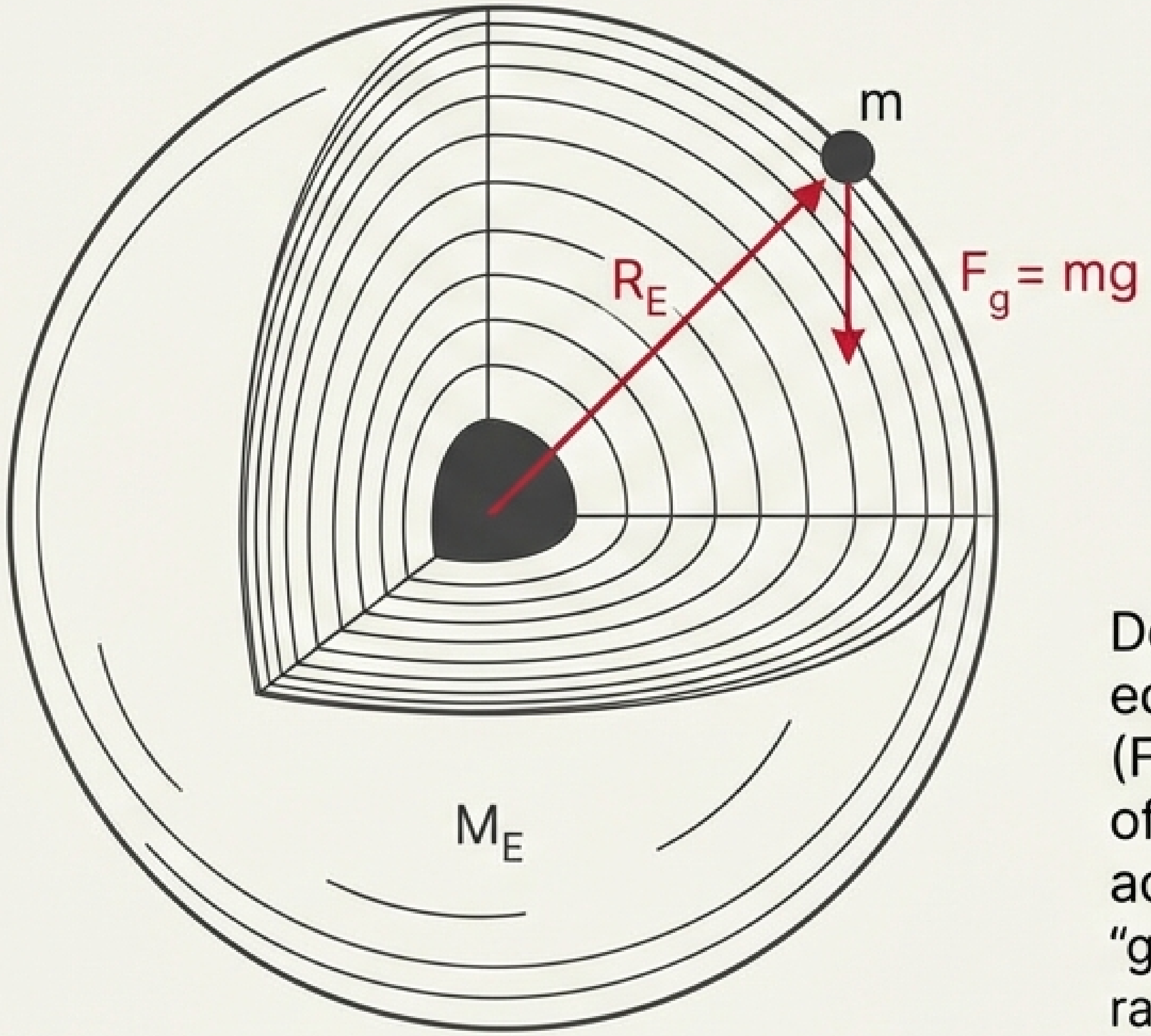
1798: Henry Cavendish “weighed the Earth” by measuring the tiny gravitational attraction between lead spheres in a torsion balance.

Key Result:
$G = 6.67 \times 10^{-11}$ N m^2/kg^2

Sidebar Note:
Equilibrium condition: Gravitational Torque = Restoring Torque of the wire ($\tau\theta$).



Acceleration Due to Gravity (g)



$$mg = G\frac{M_E \cdot m}{R_E^2}$$

$$g = \frac{G \cdot M_E}{R_E^2}$$

Deriving the local constant. By equating Newton's Second Law (F=ma) with the Universal Law of Gravitation, we find that acceleration "g" depends only "g" depends only on the mass and radius of the Earth, not the object.

Midnight Blue Derivation & Result

Equating Forces:

$$mg = G\frac{M_E \cdot m}{R_E^2}$$

Cancel "m":

$$g = \frac{G \cdot M_E}{R_E^2}$$

Constants:

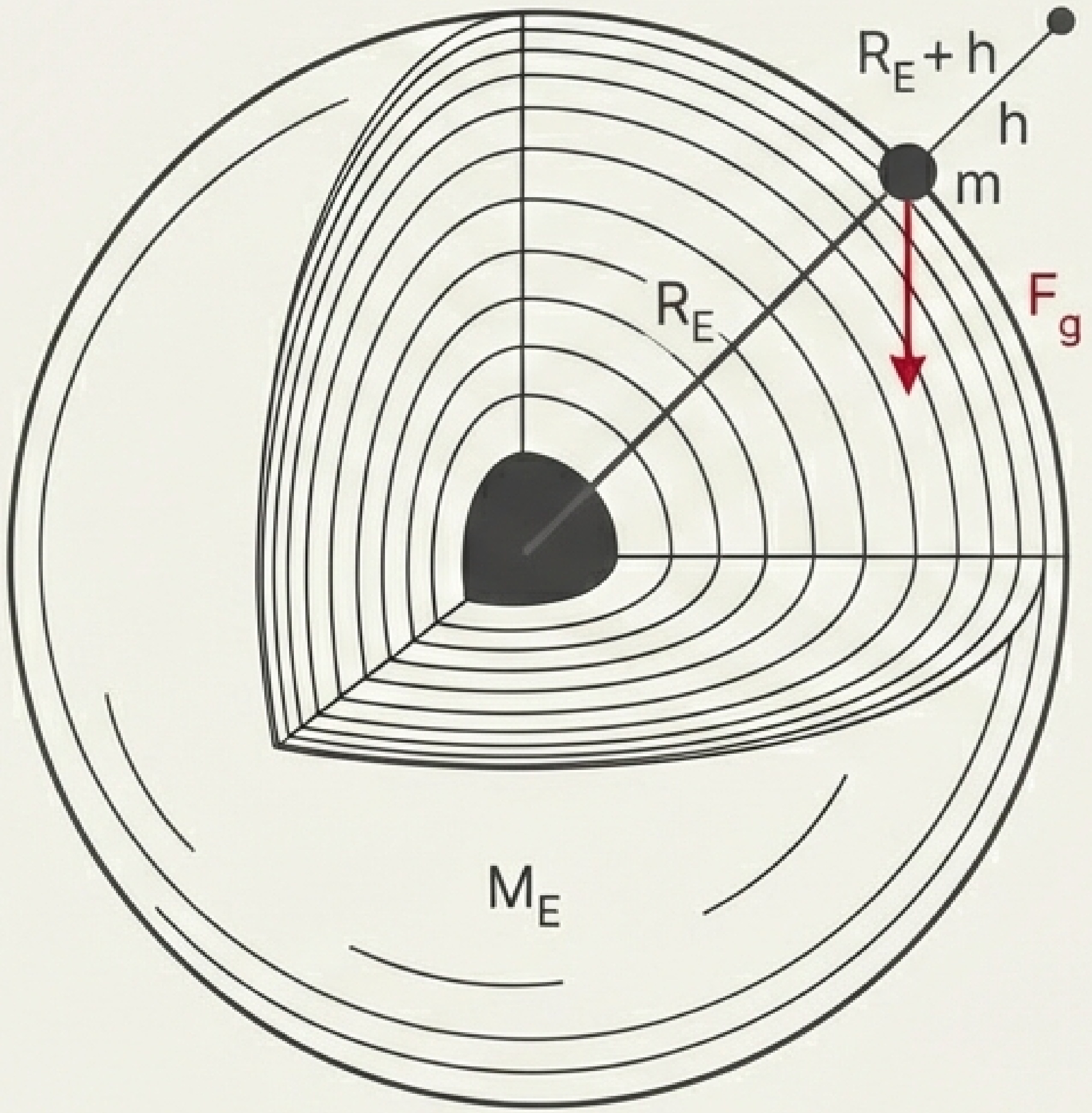
$G = 6.67 \times 10^{-11}$ N m^2/kg^2
$M_E = 5.97 \times 10^{24}$ kg
$R_E = 6.37 \times 10^{6}$ m

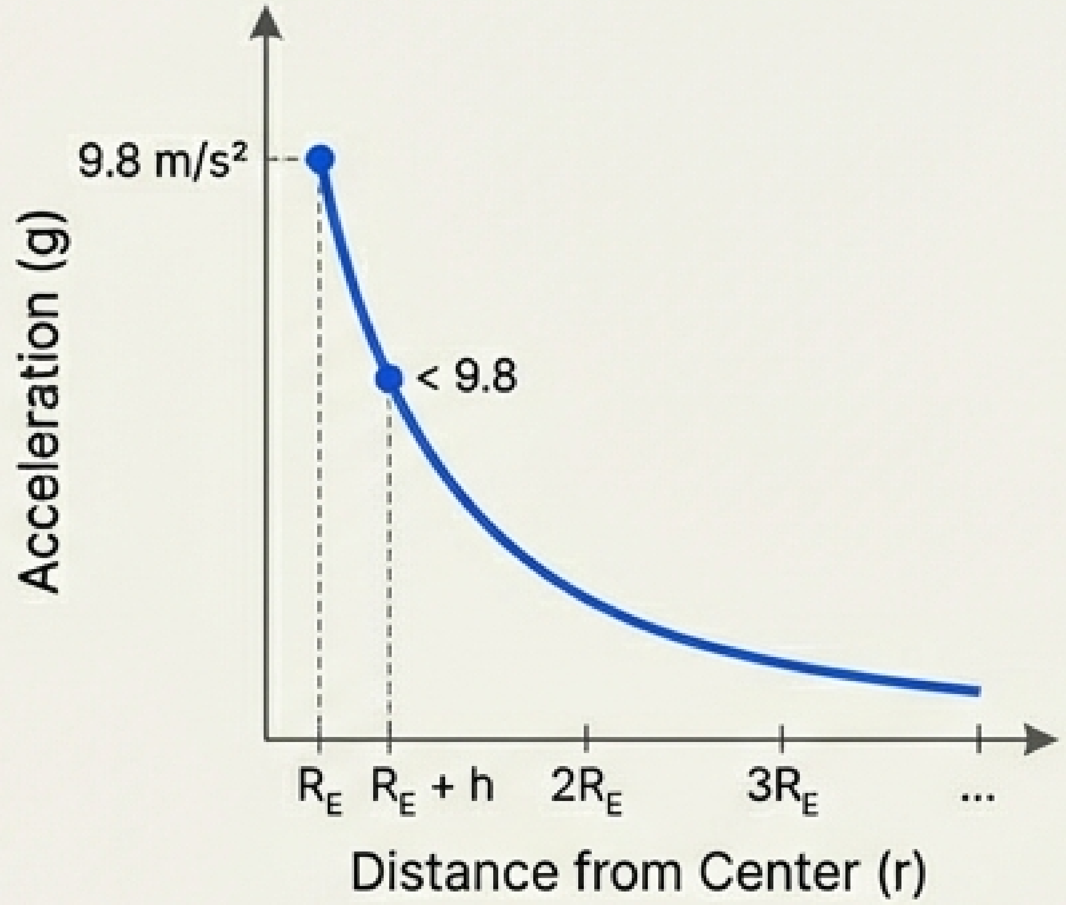
Calculation:

$\boxed{g \approx 9.81 \text{ m/s}^2}$

Scientific Crimson

Going Up: Gravity at Altitude





Gravity fades with distance. At height h, the formula becomes:

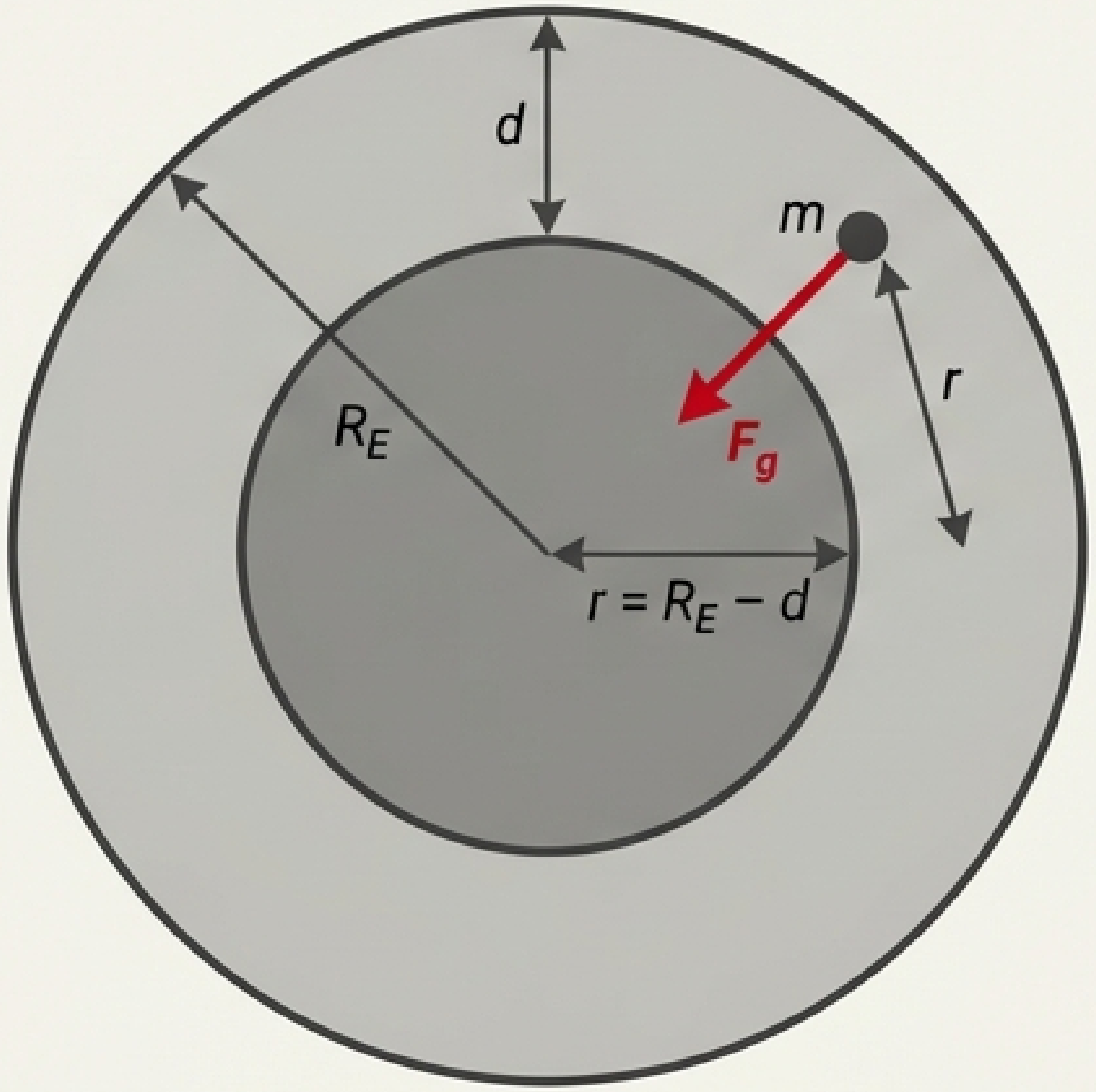
$g(h) = g\left(1 - \frac{2h}{R_E}\right)$ (Approximation for h << R)

Sidebar Insight:

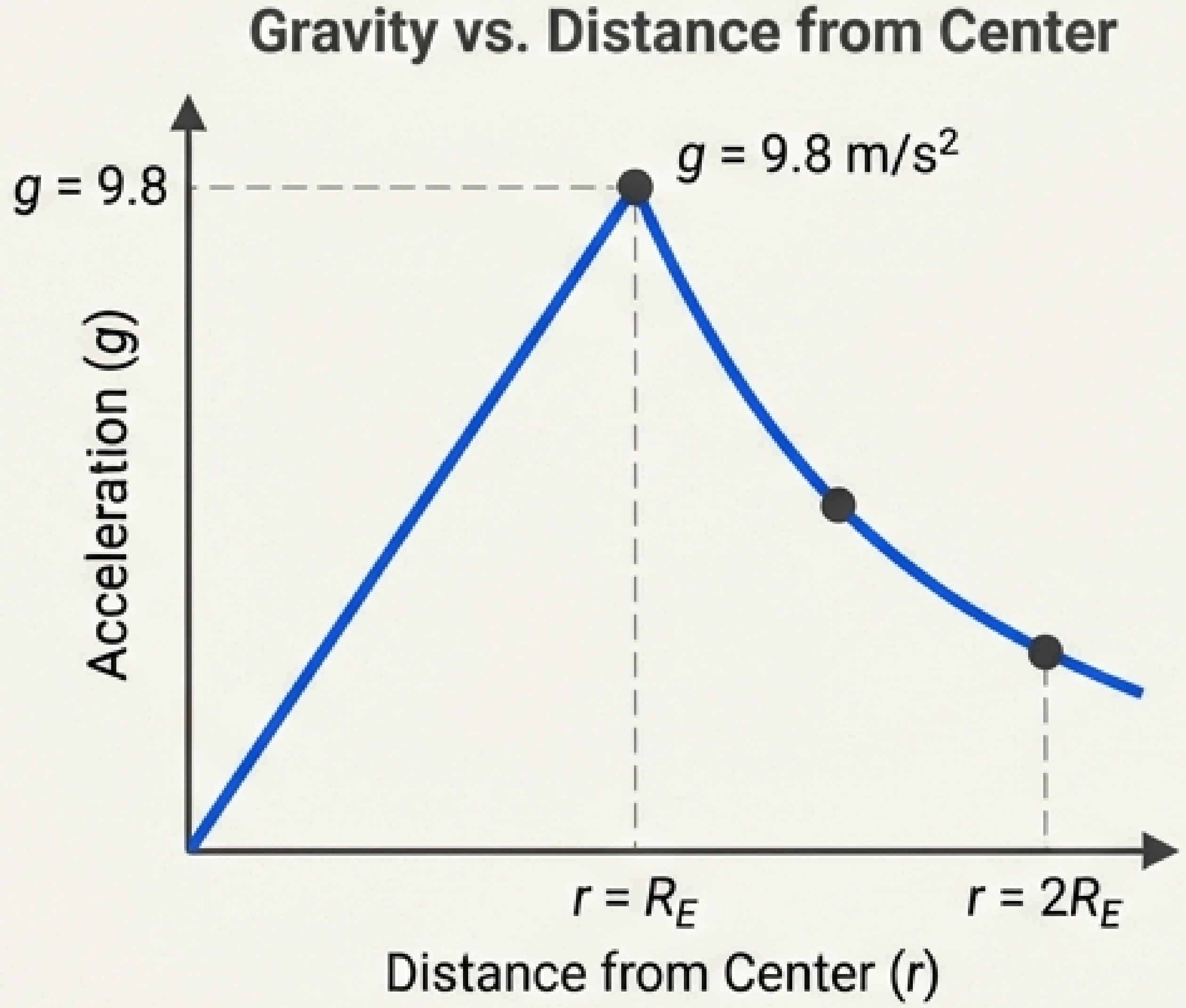
At the International Space Station (~400km), gravity is still ~90% of surface strength. Astronauts float due to free fall, not zero gravity.

NotebookLM

Going Down: Gravity at Depth



The Shell Theorem: If you are inside the Earth, the outer shell exerts zero net force. Gravity is only caused by the inner sphere.



$$g(d) = g \cdot \left(1 - \frac{d}{R_E}\right)$$

Derivation & Result

The net force inside is only due to the mass of the inner sphere (M_{inner}).

Assuming uniform density (ρ):

$$M_{\text{inner}} = \rho \cdot V_{\text{inner}} = \rho \cdot \frac{4}{3}\pi \cdot r^3$$

$$M_E = \rho \cdot V_E = \rho \cdot \frac{4}{3}\pi \cdot R_E^3$$

Using Newton's Law of Universal Gravitation for the inner sphere:

$$g(d) = G \cdot \frac{M_{\text{inner}}}{r^2}$$

$$g(d) = G \cdot \frac{\rho \cdot \frac{4}{3}\pi \cdot r^3}{r^2} = G\rho \cdot \frac{4}{3}\pi \cdot r$$

Substituting $r = R_E - d$:

$$g(d) = G\rho \cdot \frac{4}{3} \cdot \pi \cdot (R_E - d)$$

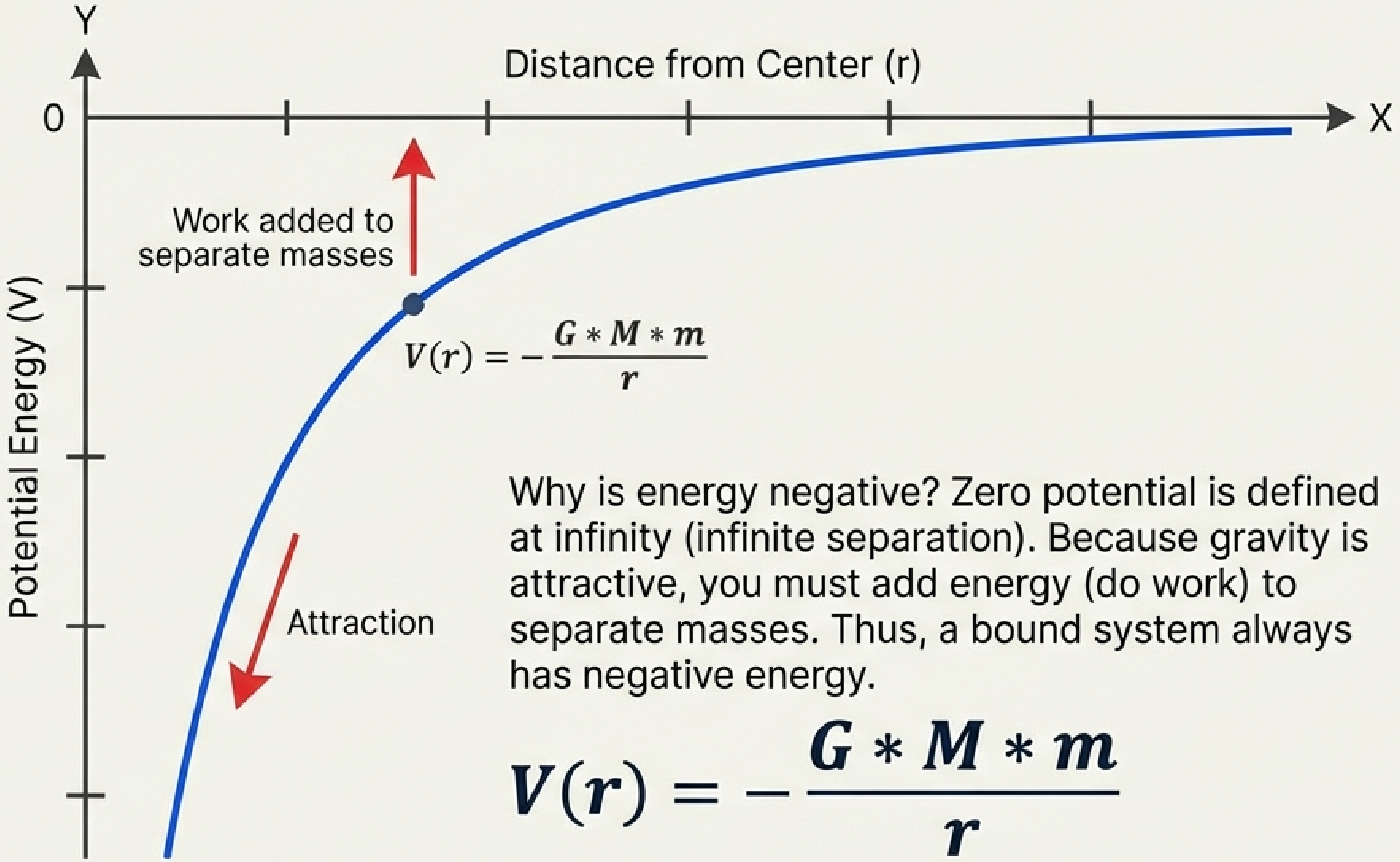
$$g(d) = g \cdot \frac{(R_E - d)}{R_E}$$

Resulting in the linear formula:

$$g(d) = g \cdot \left(1 - \frac{d}{R_E}\right)$$

Academic Editorial

Gravitational Potential Energy



Why is energy negative? Zero potential is defined at infinity (infinite separation). Because gravity is attractive, you must add energy (do work) to separate masses. Thus, a bound system always has negative energy.

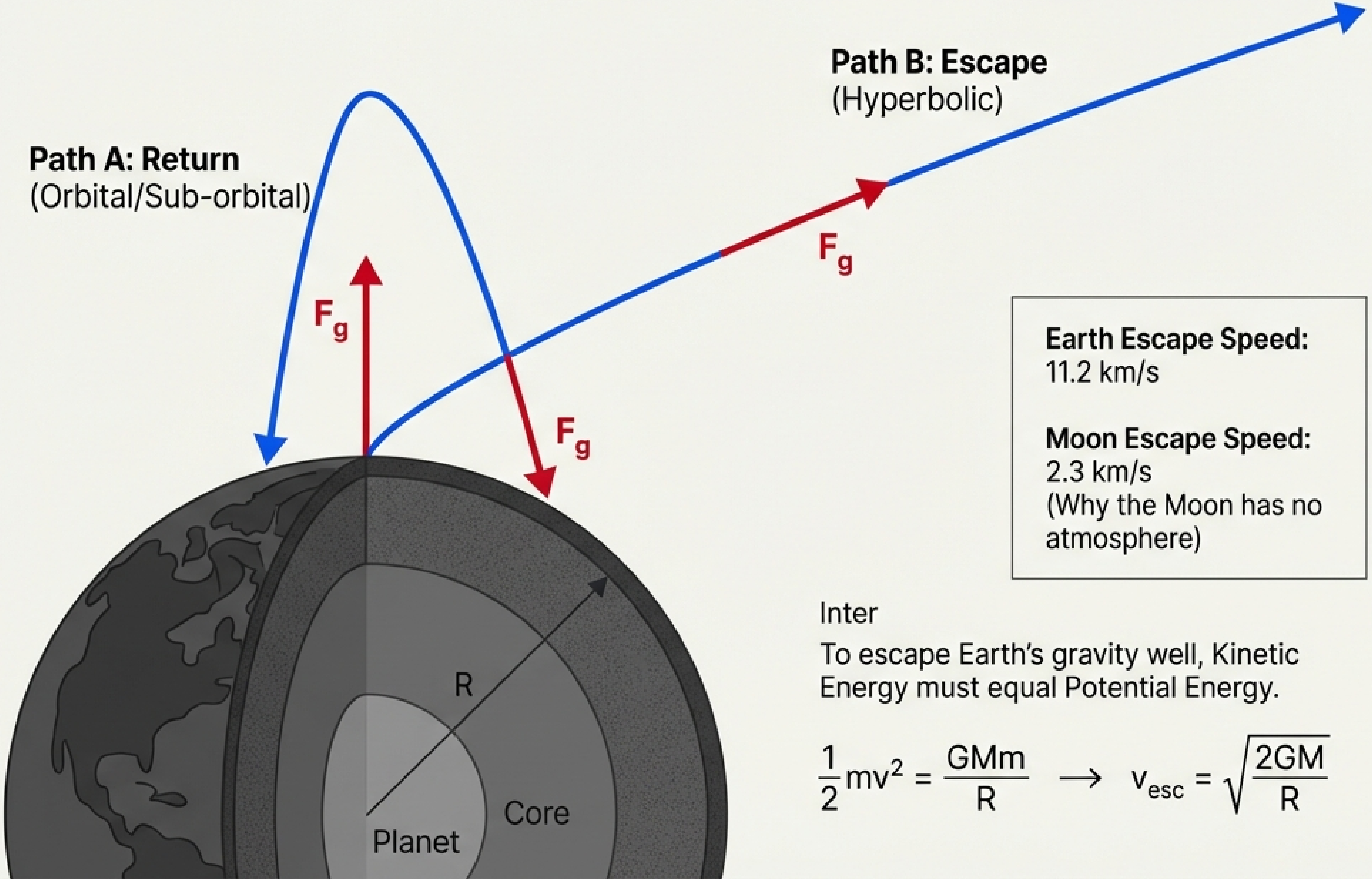
$$V(r) = -\frac{G * M * m}{r}$$

Sidebar Insight:

Near Earth's surface, this simplifies to the familiar 'mgh', but in space, we must use the universal form.

Academic Editorial

Escape Speed: Breaking the Bond



Earth Escape Speed:
11.2 km/s

Moon Escape Speed:
2.3 km/s
(Why the Moon has no atmosphere)

Inter

To escape Earth's gravity well, Kinetic Energy must equal Potential Energy.

$$\frac{1}{2}mv^2 = \frac{GMm}{R} \quad \longrightarrow \quad v_{esc} = \sqrt{\frac{2GM}{R}}$$

Derivation & Result

Inter

To escape, an object's total mechanical energy must be zero at infinity, meaning it has just enough kinetic energy to to reach infinite distance with zero velocity.

$$E_{total} = K + U = 0$$

At the surface (r=R),

$$E_{total} = \frac{1}{2}mv_{esc}{}^2 - GMm/R = 0$$

Solving for v_{esc}:

$$\frac{1}{2}mv_{esc}{}^2 = GMm/R$$

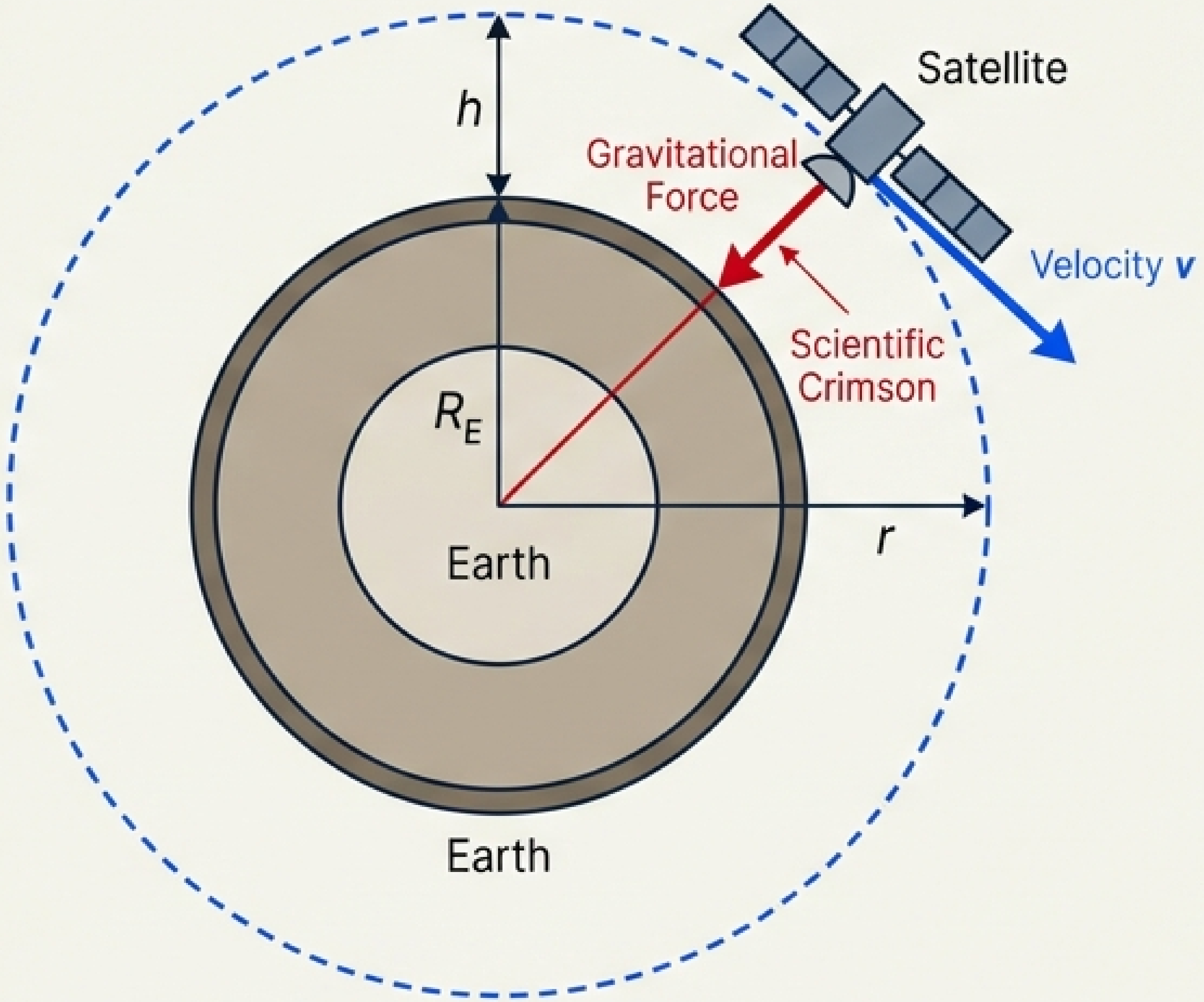
$$v_{esc}{}^2 = 2GM/R$$

$$v_{esc} = \sqrt{2GM/R}$$

The mass of the escaping object 'm' cancels out, showing that escape speed is independent of the object's mass.

Academic Editorial

The Mechanics of Earth Satellites



An orbit is a balance where gravity provides the exact centripetal force needed to keep the object turning.

Derivation & Result

Inter

Force Balance:
$mv^2/r = GMm/r^2$

Solving for v (Orbital Velocity):
$v^2 = GM/r$
v = sqrt(GM / r)

Where r = R_E + h

Key Result:
v = sqrt(GM / (R_E + h))

Note:
Velocity depends on orbit radius, not satellite mass.

Energy & Weightlessness

Energy

Total Energy (E) = Kinetic (K) + Potential (V)

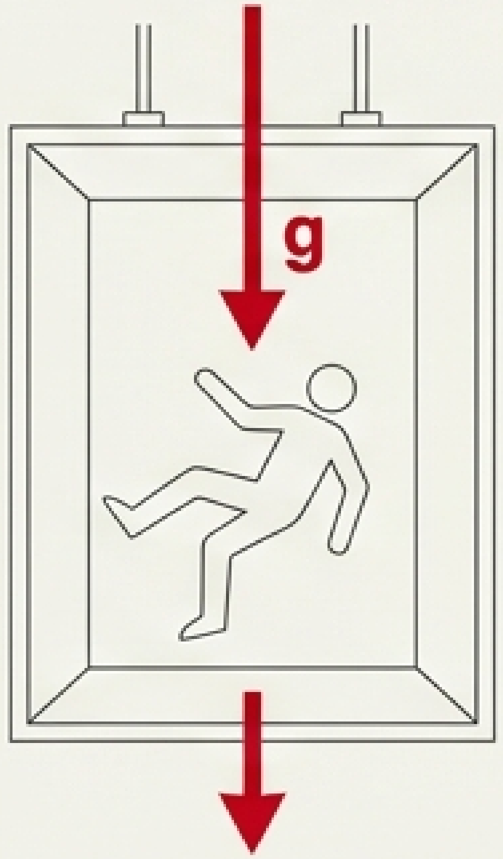
For circular orbits: $E = -\frac{GMm}{2r}$

The energy is negative, meaning the satellite is trapped in the well.

Weightlessness

Misconception: "There is no gravity in space".

Reality: Gravity at the ISS is strong (~$8.9\ m/s^2$). Astronauts feel weightless because they are in constant Free Fall. The floor of the spaceship is falling at the same rate as the astronaut, so there is no normal force to push back against their feet.



Derivation & Result

Orbital Energy Derivation:
Total Energy $E = K + V$

For circular orbit, $v^2 = GM/r$, so
$K = \frac{1}{2}mv^2 = \frac{1}{2}m(GM/r) = GMm/2r$

Potential Energy $V = -GMm/r$

$E = K + V = GMm/2r - GMm/r = -GMm/2r$

The negative sign indicates a bound state.

Free Fall Principle:
An astronaut in orbit is in continuous free fall around the planet, experiencing only the gravitational force ($F_g = mg$). There is no normal force from a supporting surface, resulting in the sensation of weightlessness.

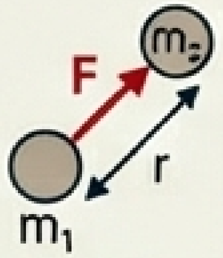
Academic Editorial

Summary & Key Formulas

Universal Law

$$F = G\frac{m_1 m_2}{r^2}$$

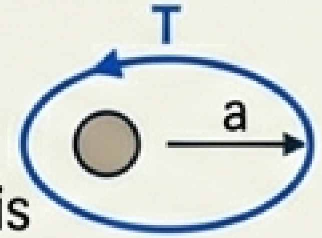
Constant:
G = 6.67e-11 N·m²/kg²



Kepler's 3rd Law

$$T^2 \propto a^3$$

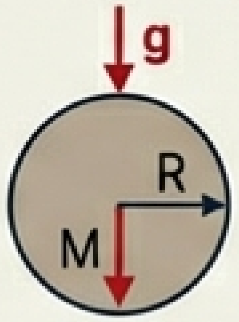
T = orbital period
a = semi-major axis



Surface Gravity

$$g = \frac{GM}{R^2}$$

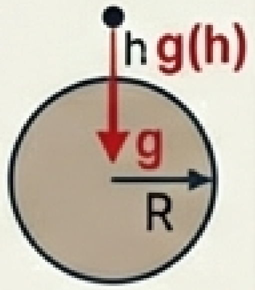
G = Gravitational constant
M = mass of body
R = radius of body



Gravity at Altitude

$$g(h) \approx g\left(1 - \frac{2h}{R}\right)$$

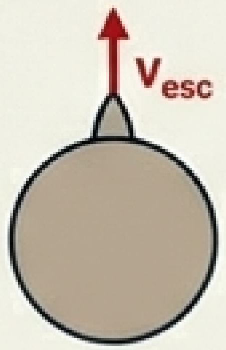
h = altitude above surface
R = radius of body
g = surface gravity



Escape Speed

$$v_{\text{esc}} = \sqrt{2gR}$$

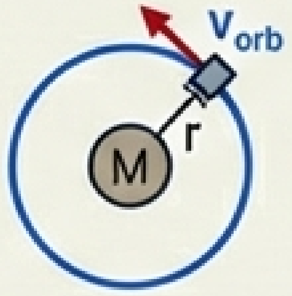
g = surface gravity
R = radius of body
sqrt = square root



Orbital Speed

$$v_{\text{orb}} = \sqrt{\frac{GM}{r}}$$

M = mass of central body
r = orbital radius
sqrt = square root



Gravity is the architect of the universe. It binds us to the Earth, holds the atmosphere, directs the moon, and keeps the planets in their eternal dance around the sun.

Derivation & Result

Orbital Energy Derivation:

$g = \frac{GM}{R^2}$

For circular orbit,
$v^2 = \frac{1}{2}mv^2 = \frac{1}{2}GM/r$, so
$K = \frac{1}{2}mv^2 = \frac{1}{2}GMm/r$

For circular orbit Derivation:

$K_{\text{ift}} = V = \frac{1}{2}Gm/r$

Potential Energy V = -GMm/r

E = K+V = GMm/2r - GMm/r =-GMm/2r

$$v_{\text{orb}} = \frac{\sqrt{GM}}{2r}$$

$$v_{\text{orb}} = \sqrt{GM - \sqrt{\frac{GM}{r}}}$$

*There as the architect of the universe. It binds us to the Earth, holds the planets in their eternal dance around the sun.